

MEAPER

Modular Electroactive Actuator with Passive Elastic Restoration

Egor Fenko, 25 aug 2026

Abstract

Actuators are commonly divided into two groups: soft and rigid. While soft robotics offers high adaptability, these systems often lack the structural stability and precision of rigid mechanisms. To address this trade-off, this project presents a novel modular actuator that bridges the gap between soft flexibility and rigid controllability. The system achieves this by utilizing a network of coupled cells, each of which has a controllable range of contraction. Since they are formed in a closed manifold structure, together they produce complex motion.

For validation purposes, the system was built from 3D-printed cells, actuated by nickel-titanium shape memory alloy wires (NiTi). To achieve precise contraction, a specialized pulse-width modulation on electroactive material was employed and were used torque restoring springs for elastic restoration. This paper presents kinematic modeling, mechanical design, and calibration of the components.

Finally, the system was evaluated using a programmed kinematic control pipeline. Key metrics such as spatial precision and overall working volume were measured. Results show that the architecture is highly scalable in both motion and control; adding a single modular layer expands the end-effector's reachable volume by a factor of 12.45. Furthermore, computational cost grows linearly with the number of layers, and positional errors do not propagate between sequential calculations, demonstrating the actuator's viability for hybrid soft-rigid manipulation.

Introduction

The main controllability challenge in soft robotics is building a system that handles infinite degrees of freedom (a number of independent directions of motion a robot has). Material that is used in such actuators, that is soft and elastic, is most accurately modeled using partial differential equations. Solving them for material kinematics is computationally heavy and it is practically impossible to use this method for live control loop deployment without approximation techniques [1,2], such as neural network approximation of differential equation solution or simplification of the structure itself by reducing it to discrete building blocks (volumetric Finite Elements methods), some even try reinforcement learning to solve multimodality [4], which requires heavy training and data gathering. All these approaches try to simplify complex structure algorithmically. In contradiction this paper suggests to reduce complexity of the mechanical structure itself to a level that its inverse kinematics computation will be easily deployed on low budget high end microcontrollers.

The approach of discretization of continuous surfaces into polygonal structure [3] is not new and used both in graphics and material physics. To bypass the tradeoff between modeling accuracy and computational simplicity, MEAPER shifts the discretization from the software to the hardware, employing an inherently discrete polygonal cellular architecture.

This implementation of MEAPER architecture and control pipeline uses SMA (Shape Memory Alloy) and restoring torque springs to mimic elasticity of a single structural unit of soft actuator and it inherits rigid closed manifold structure, which reduces its degrees of freedom and makes it mechanically stable and stiff against unwanted deformations. In addition to that, this project introduces a way of sequential control for uniform closed manifold structure of general length instead of using a single heavy model that produces a solution for the whole structure, which is extremely valuable for microcontrollers, due to their memory space limitations.

Related work

In Kinematics for an Actuated Flexible n-Manifold [5], Dr. Oded Medina develops forward and inverse kinematics (finding parameters of the structure that satisfy some specific geometrical condition, usually of end effector) calculations for redundant, flexible manifold mechanisms. This project advances that approach by combining a neural network for initial predictions with a generic second-order numerical solver to resolve the inverse kinematics. Similar to [5], this work resolves the inherent geometric redundancy of the structure through energy minimization. Furthermore, this project utilizes identical SMA components [8], as [5] demonstrates SMA to be a highly effective actuating material for manifold architectures.

A secondary challenge in resolving structural redundancy involves modeling the distribution of possible kinematic solutions. The computational approach used in this project was inspired by Neural Inverse Kinematics [12], which introduces a Gaussian Mixture Model with hypernetwork guidance to model the positional distribution of joint angles. This project adapts that architecture by integrating a Fourier feature mapping layer. This update is critical in order to overcome the spectral bias of neural networks, allowing the model to accurately capture the high-frequency spatial coordinate data required for precise end-effector placement.

Approach

1. Background

This project focuses on a specific geometric structure chosen for an actuator: a closed polytope. This representation is highly convenient for modeling because it restricts the available degrees of freedom. Unlike open surfaces (manifolds with boundary), a closed polytope is a compact domain. This lack of open boundaries ensures that solving kinematic and geometric

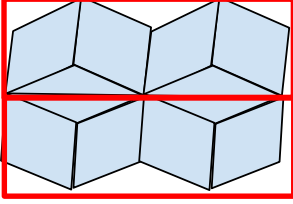


Figure 1 - example of division of the 3d structure into layers

constraints over the structure remains mathematically and numerically stable. For simplicity of modeling rhombuses polygons, connected together by edges, will be used. The decision of choosing rhombuses was made in favour of simpler fabrication. Rigid edges create constraint over the shape and the only thing that can change is the inner angle of the rhombus, which due to its geometry solves the entire figure (given one inner angle we know all other angles in its plane and both diagonals). Moreover when we add Nickel Titanium wire as a diagonal element the most stable geometrical structure is obtained - triangle.

2. Inverse and forward kinematics

Inverse kinematics in this case would be formulated as follows: given end effector position $(\bar{x}, \bar{y}, \bar{z})$ and direction $(\bar{x}, \bar{y}, \bar{z})_{\perp}$, what should be the angles of all rhombuses in the structure.

Closed polytope, made of rhombuses can be easily divided into layers, for example as one can see on Figure 1, we can divide the structure where horizontal diagonals of middle rhombuses pass. Important to notice that it is not guaranteed that in closed shape nodes that are connected by horizontal diagonals will lie in the same plane. The end effector in this case will be the centroid (mean along each axis of points in the layer) of the last layer. Its direction - normal of surface that includes all nodes within minimal margin. We will also fix the base of this structure and assume that we know coordinates of layer 0 nodes.

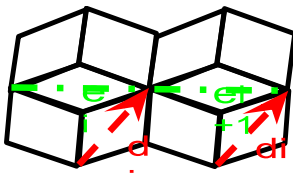


Figure 2 - parametrization of diagonals, e is used for edges and d for diagonals

Given how coupled the structure is, a suitable forward kinematics problem would be following: given all vertical distances $(d_1, d_2, \dots, d_n)$ and connecting them horizontal edges $(e_1, e_2, \dots, e_n)$ (Figure 2) what is the position of all nodes in structure, given layer 0 coordinates $(\bar{x}_1, \bar{x}_2, \dots, \bar{x}_n)_0$. To solve this, one should apply an iterative optimization algorithm. Given previous layer nodes coordinates $(\bar{x}_1, \bar{x}_2, \dots, \bar{x}_n)_i$ and diagonals $(d_1, d_2, \dots, d_n)$ and

edges $(e_1, e_2, \dots, e_n)$, next layer nodes $(\bar{x}_1, \bar{x}_2, \dots, \bar{x}_n)_{i+1}$ would lie on surface of sphere that centered at point $\bar{x}_j$ with radius d_j . So the optimization problem becomes:

$$\min_{\{\bar{d}_k\}_{k=0}^n} \frac{1}{n} \sum_{k=0}^{n-1} (e_k - ||(\bar{x}_k + \bar{d}_k) - (\bar{x}_{(k+1) \bmod n} + \bar{d}_{(k+1) \bmod n})||)^2, \text{ minimization of the}$$

difference between real edges and edges that are formed between coordinates created by taking step $\bar{d}_k$ from point $\bar{x}_k$. As we already know d_k , we just need to find its direction. This can be solved

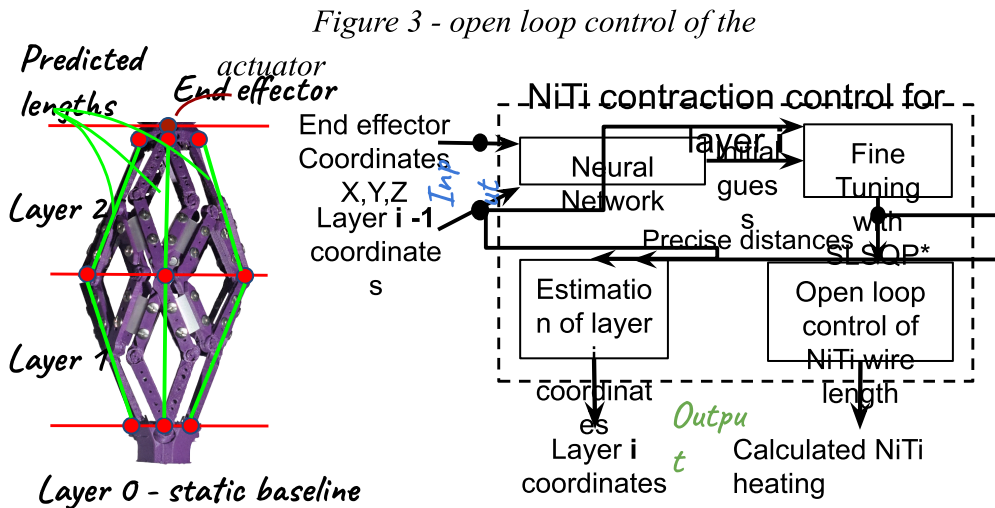
in spherical coordinates by estimating angles $\{\theta, \phi\}_{k=0}^n$. This can not be solved from geometry. The problem does not have a single solution, it is underconstrained and lacks information. For each node we have 2 variables to find (total 2n), for n nodes there are only n constraints

$e_k = ||(\bar{x}_k + \bar{d}_k) - (\bar{x}_{(k+1) \bmod n} + \bar{d}_{(k+1) \bmod n})||$. In a closed loop there are n edges between n nodes. Which would require additional n constraints to make this potentially solvable analytically. However that would make the kinematics determined and rigid and the model will not be useful for modeling the actuator we are about to build.

This problem is still ambiguous and non convex, because even if we look for a solution in the upper semisphere, there are still 2 chiral configurations that give exactly the same solution to the formulated optimization problem. That is why, when one solves this iteratively with Gauss Newton optimization step, it is important initializing the angles with values, same as on the previous layer, leading to convergence towards a single minimum, preventing structure from being twisted.

3. Control loop

Most inverse kinematics methods use conditioned calculation of next state given previous. This method suggests open loop control that eliminates degradation of control over time by making it state independent. On Figure 3 one can see how iteratively solution could be found by passing i layer nodes $(\bar{x}_1, \bar{x}_2, \dots, \bar{x}_n)_i$, target centroid of the next layer $(\bar{x}, \bar{y}, \bar{z})_{i+1}$, and its direction $(\bar{x}, \bar{y}, \bar{z})_{\perp, i+1}$ into the neural network to get noisy guesses of diagonals $(d_1, d_2, \dots, d_n)_i$ and edges $(e_1, e_2, \dots, e_n)_i$. Then guess is optimized by second order optimizer SLSQP [13, 14]. So the lower boundary of the



neural network performance is to guarantee convergence in all cases for second order optimizer, and the upper is to make the guess as close as possible due to the high computational cost of second order optimization. Then polished by SLSQP diagonals and edges are inserted into the same Gauss Newton optimization algorithm described above, so that we will get reliable estimation of parameters for the next iteration.

We already assumed that layer 0 nodes position is known, so we can reliably perform this operation for an unlimited number of layers leaving the only dependency of the neural network on per layer local distributions of nodes.

4. Actuation

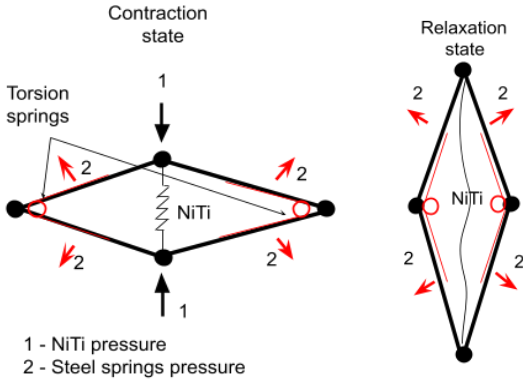


Figure 4

From the same control loop predicted edges and diagonals are directly mapped to SMA wires that pass through diagonals of each rhombus of the structure. The target distances are translated into voltage regimens over NiTi (Nickel Titanium alloy) wires and applied at the same step, which enforces deformation towards position that guarantees end effector correct positioning because the electrical current that runs through the NiTi wire enforces

contraction. For relaxation of the structure, which results in an increase of diagonal length, we stop pushing power into the wire and opposing torque springs restore the initial shape of the rhombus. This elastic restoration makes this design energy efficient. The reason behind that is that compensation of opposing force comes not with larger power consumption by NiTi, but with larger wire diameter, which indeed requires more heat for initial heat up, however it also accumulates the heat better, so it requires lower frequency pulses which result in overall gain in energy preservation.

Methods

1. Dataset generation

Using this method one can do Monte-Carlo random sampling as paper [12] suggests. In this specific case all diagonals and edges were generated within $\sim 15\%$ standard deviation, leaving some room for convergence. Those samples that converged successfully were kept and used as training dataset.

2. Neural Network

The base of all architectures that were tested was IKNet published in [12]. The reason is simple - this architecture with GMM (Gaussian Mixture Model) prevents averaging in multimodal scenarios. Usage of networks that are not designed for multimodality will result in averaging structure configuration over same end effector position, which is not desired, because in non linear systems average of inputs does not lead to average of outputs, so this will result in an inaccurate estimation.

Another important point to consider is spatial frequency of input data. The architecture is extremely coupled so small changes in diagonal will just disappear if we explicitly do not state that

the network should capture high spatial frequencies. The high frequency change in adjacent diagonals can be amplified to an entirely different end effector position. Neural networks tend to learn low frequencies first [9] and just interpolate high frequency dependencies along with noise. The data used in this training is synthetic, SNR is very high, so a Fourier Feature map will not overfit over noisy data.. That is why the architecture of the network includes a Fourier Feature map [10]. It helps to eliminate spectral bias of the network and learn high grain features. Below you can see the forward autoregressive pass that was used in this project for sequential prediction of manifold layers positioning.

For training the network uses combined loss:

$$L = NLL + \alpha \text{ Power Penalty}$$

Where the power penalty minimizes diagonal lengths and maximizes edges. The reason is behind placement of NiTi wires that are heated. We want our target configuration of edges and diagonals to utilize as few NiTi contracted wires as possible to save energy. Heating power is directly proportional to wire length, so we can simply include non squared length terms inside the loss function in order to minimize power loss.

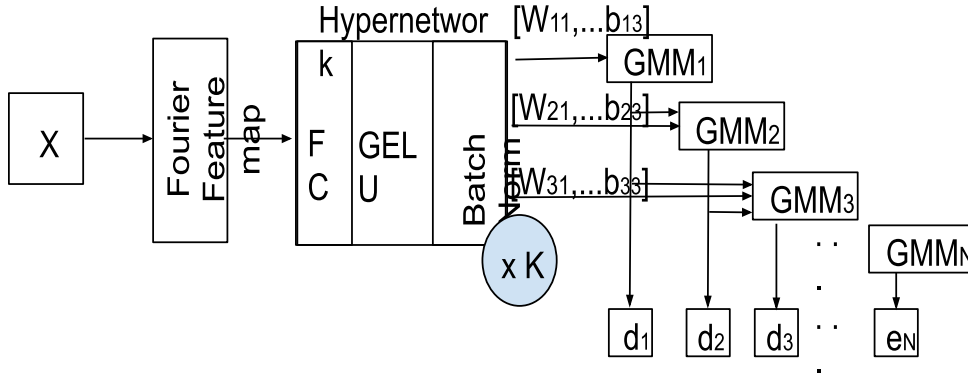


Figure 5

3. SLSQP

This second order optimization algorithm [14], as all second order optimization algorithms require very close initialization. In results there is no numerical data in the results section that covers SLSQP convergence, because models were optimized until all values in the working area will not converge with SLSQP, the time latency that Figure 10 shows already includes SLSQP. So the model that was closest to its true position was optimized faster. Hyperparameters for SLSQP such as maximal iterations count, were chosen so that 80% of samples in the test batch (taken from validation dataset) will converge before reaching that limit. It manages to optimize every target position to 0.01 mm precision (relative error $\ll 1\%$), and generally speaking this precision is not needed in most applications, that is why computation speed can be increased with the cost of accuracy.

Target square programming problem that SLSQP optimizes is:

$$\min_{L_i} \sum_{j=1}^M (||L_{i,j} - L_{i-1,j}||_2^2 + ||d_{nm}||_2^2 + ||E_{i,j}||_2^2 - e_{nm}||_2^2) + \gamma ||E_{i,x} \times E_{i-1,y} - E_{i,y} \times E_{i-1,x}||_2^2$$

s. t.

$$|| < L_i > - Target Centroid_i || = 0 \quad \text{Aligning centroids to get correct position}$$

Feasibility constraints according to rigid parts of actuator: $\forall j \in [M]$

$$\min val \leq ||L_{i,j} - L_{i-1,j}|| \leq \max val$$

$$\min val \leq ||E_{i,j}|| \leq \max val$$

where $L_{i,j}$ is coordinate of point j in layer i, d_{mn} , e_{mn} are diagonals and edges that are predicted by neural network that correspond to point j, E_i is $L_i - PL_i$, where P is rotation by one matrix.

- $||E_{i,x} \times E_{i-1,y} - E_{i,y} \times E_{i-1,x}||_2^2$ is twist penalty between layers, so that rigid mechanical structure will not break
- $||L_{i,j} - L_{i-1,j}||_2 - d_{mn}||_2^2$ forces to be close to diagonal predictions done by neural network
- $||E_{i,j}||_2 - e_{mn}||_2^2$ forces to be close to edge predictions done by neural network

M - number of points in a layer.

The last layer has an additional term that also forces edges to be an exact square, because of the rigid top of the structure. By optimizing this problem over layer i coordinates we practically shift next layer coordinates to fit both to guessed configuration by nn and to make the final solution feasible.

Target centroid can be estimated using different methods. In order to keep energy and deformation equal across layers, centroids were calculated as equidistant points across the vector that connects the end effector and base centroid. This however does not guarantee that precalculated centroids will be reachable for layer i, however we still want to minimize distance between them. This leaves some space for further tests by setting non uniform centroid positioning, along a custom curve.

4. NiTi voltage regimen

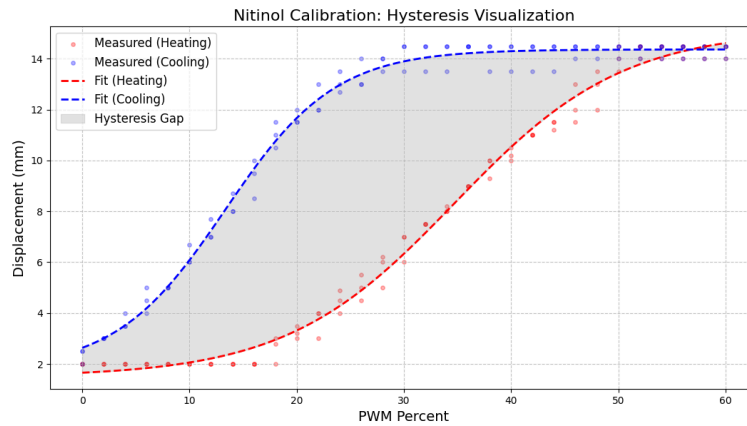


Figure 6 - hysteresis loop of Flexinol [7], blue curve corresponds to cooling from austenite phase and red curve corresponds to heating from martensite phase, manual measurement by sampling and fitting to sigmoid function

Nickel Titanium has the following property - when it heats up it becomes “elastic” (austenite phase) and when it cools down it becomes “plastic” (martensite phase). This property allows it to be shaped as a memory alloy. Because the austenite shape of crystalline can be pretrained by heating the material to 450-550 degrees Celsius. This allows training the wire into spring-like shape, which

can be used as a mechanical amplifier of contraction within alloy heating, which is a general property of metals (with no shape deformation contraction reaches only up to 5% of total length).

The trained material contraction can be mapped to a power regimen that is used to heat it up. This project utilized the known PWM approach [15], which is a superior method for SMA actuation in robotics. It delivers energy in pulses which prevents overheating the wire (that can cause it losing its shape memory properties) and in the same time tuned PWM can result in stable enough temperature. The correct power regimen should reach physical limits of the rhombus frame and keep it responsive, one should search which PWM (Pulse Width Modulation) delivers the maximal current required for full contraction of the SMA according to documentation, which will define the upper limit. There are two options to choose a lower limit voltage regimen. First one is to drop it to 0, this will preserve maximum energy when the cell is relaxed, then when contraction is needed a high energy pulse that exceeds requirement should be sent into the material, which will cause rapid preheat and then correct voltage should be applied according to mapping. This is a good approach when actuation of structure is sparse, which is true for some geometries, or if control loop was deliberately pretrained to produce sparse diagonal activations. This is not the case here - that is why here PWM % drops to the highest level that drops the temperature of SMA enough to let torque springs compress it. By that, material is always ready to be actuated and does not require spikes that can overshoot.

The other challenge with NiTi is that it has a non linear hysteresis loop, that fits to a couple of sigmoids ($\tanh$) [11]. This means that the curve of heating does not equal the curve of cooling. If one will look at Figure 6, one can see that heating from a relaxed state differs from cooling from a contracted state. That is also true for any point inside the loop, which results in a continuous set of non linear functions inside the hysteresis loop. This can be accurately modeled as a state space model, however that is too complex for fast calculation. An alternative that was used in this project is a fixed number of states and linear approximation of inner hysteresis loops, simultaneously wiping off state memory by low frequency jitter.

5. Modeling for validation

A model for an actuator was designed and printed on a 3D printer (Prusa MK3). The unit cell consists of following rigid edges as one can observe on Figure 7.

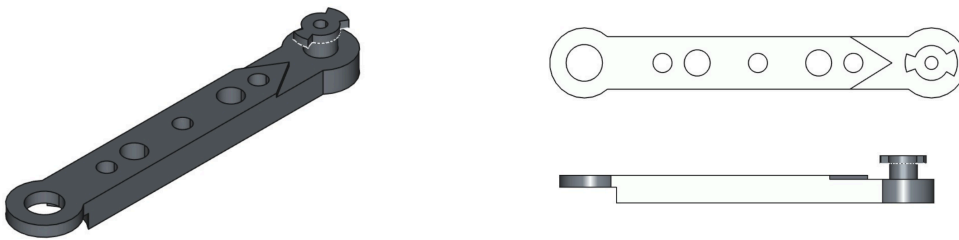


Figure 7 - rigid edge design for rhombus unit cell

Each edge is 7cm long, which is limited by the resolution of the printer that was used. At the corner of each edge there is a point where NiTi wire can be attached. Along its length there are other connections that allow it to attach torque springs and elastic interconnections between rhombuses. Elastic material between adjacent cells is essential, because it guarantees that there is a space for motion that fits the model that was developed earlier. The space between cells can be modeled as

virtual polygons that are tightly coupled with those we can control, that is why their existence does not change an optimization problem.

Materials

Actuation

Nickel Titanium alloy was chosen as an actuation material, however given structure and inverse kinematics algorithm any type of linear/rotational actuation can be used to change rhombuses geometry. This alloy has proven itself as a durable and easily trainable material found in many robotics researches including [5]. Specifically Flexinol® 0.012" diameter [7]. It requires 1.5A current for full contraction, that is why circuitry for its control required decoupling high voltages from low, to keep the microcontroller safe.

Hardware

The microcontroller that was used is ESP32-S3 [6] allows vector operations and has 2 cores, which makes it a good fit for this application.

Heating circuit

PWM signal was generated on one of its outputs, then passed through optocoupler and to the ground. On the other part of optocoupler there is high voltage DC signal that enters optocoupler from the transmitter side, one to the ground through voltage divider, divided voltage goes to high voltage transistor gate, its source and drain are connected to same high power circuit passing signal through NiTi wire. An inverted diode on the drain helps to prevent voltage spikes from entering the transistor and heating it up. Figure 8 shows the exact circuit that was used in this experiment to heat up NiTi. An air fan cooler was used to force temperature reduction during relaxation. Multiplexer allows the reuse of the same schematics over each diagonal module in parallel.

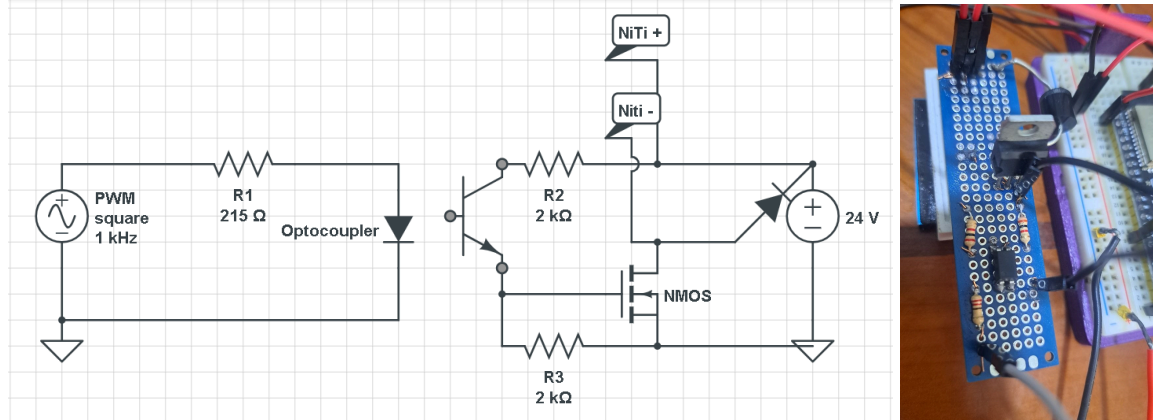


Figure 8

Libraries

SciPy [13] was used as a source of SLSQP. There are other implementations, however it was not the target to check SLSQP performance, but neural network and controllability in general so the choice was arbitrary.

Prototyping

Printed details were printed with PETG [8] because it has relatively high glass transition temperature (68 degrees Celsius) and as a result was repeatedly used in experiments with Nickel Titanium alloy with no deformations.

Software

The simulation of control pipeline application can be found on projects Github link, listed at the title page. There is no requirement on PC, just some basic libraries need installation, such as numpy, matplotlib or pytorch. Exact list of them one can find and installed with requirements file in project directory as: `pip install -r requirements.txt`

Evaluation and Verification

Evaluation metrics:

- Flexibility and scalability

This is in the first place a hybrid actuator that mimics soft actuator kinematics. That is why one of the most important metrics is volume covered by the end effector and how it improves with additional layers. Results of Monte Carlo random sampling and fitting to Gaussian sphere showed that initial volume barely covers the base of an actuator, being projected on it.

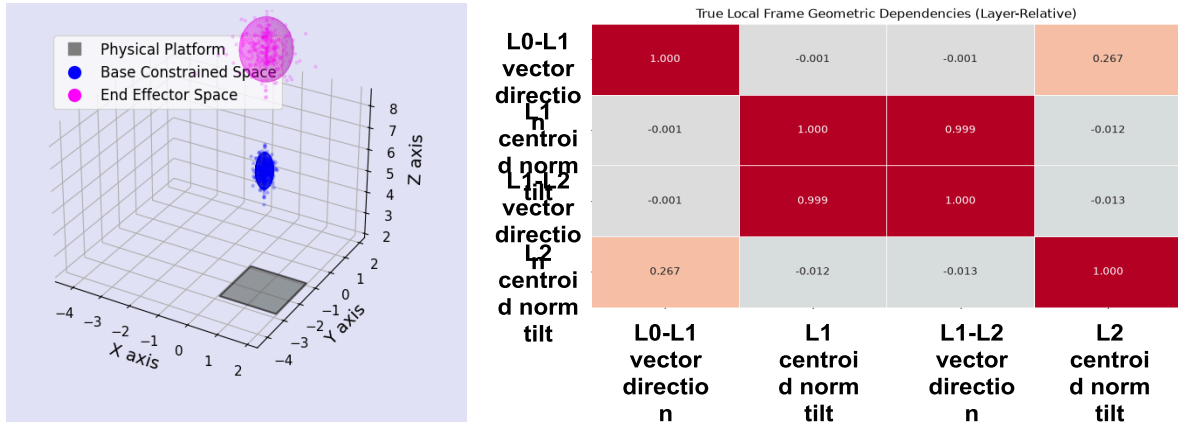


Figure 9 - (on the left) how the workspace of the centroid changes with additional layers. In the blue centroid of the first layer. In the pink centroid of the second layer.

(on the right) table of correlation between different bendings in the system - flexibility & coupling

Results show that Adding a single additional modular layer amplifies the end-effector's reachable volume by a factor of 12.45, in addition to that table on the right shows that end effector is completely decoupled from previous layer, acting like a ball joint, slightly dependent on first layer direction, which is clearly a consequence of limited motion range in general. This shows us that the structure indeed resembles some sort of soft material rather than rigid joint.

- Performance of different models

Figure 10 shows that sequential layer estimation is superior for this specific architecture, because as a number of layers grows, the capacity of IKNet should also grow and in addition to that sampling becomes less and less reliable as the number of degrees of freedom grows. Sequential models each time are conditioned on fixed, polished centroid, estimated by reliable gradient based optimization. That is why error does not propagate, but is being fixed every next layer, so the only error we have is mean error of single layer estimation.

Each architecture parameters such as number of layers, frequencies, width were chosen such that architecture would show its best performance over a training set and then chosen architecture

was trained to convergence, this does not guarantee that this performance is best possible for each shown architecture, however it gives rough estimation that clearly shows trend.

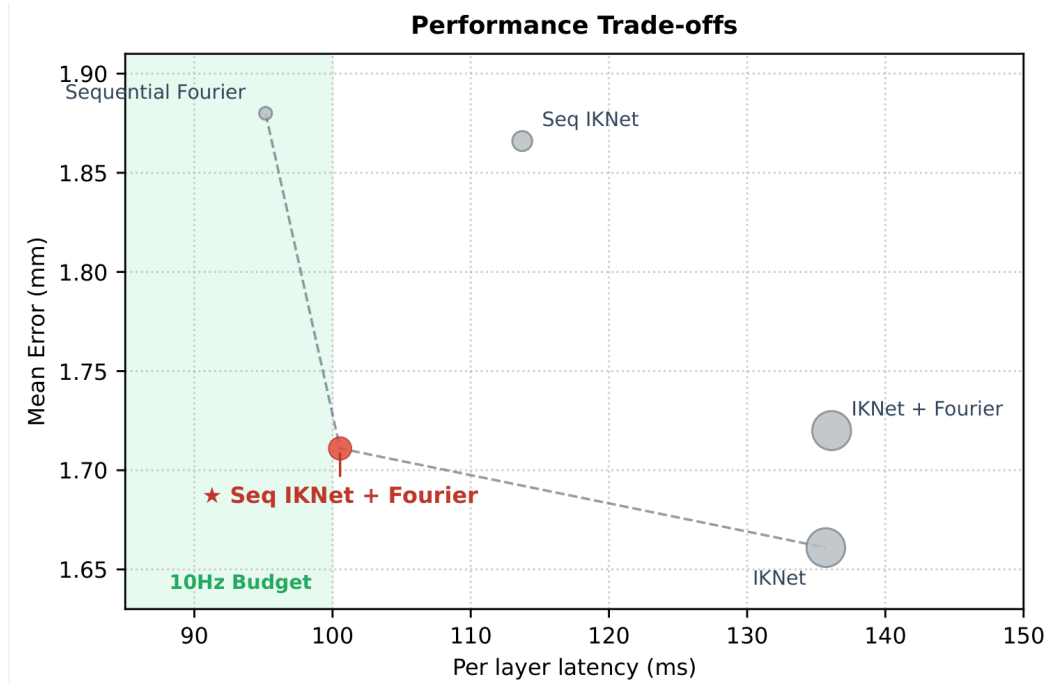


Figure 10 - Comparison between different prediction methods. The size of bubbles demonstrates the physical size of the best performing model. left-down better.

Model name	Estimation strategy	Fourier Features
Sequential Fourier *	Layer-wise sequential	+
Seq IKNet	Layer-wise sequential	-
IKNet	Global (all parameters at once)	-
IKNet + Fourier	Global (all parameters at once)	+
Seq IKNet + Fourier	Layer-wise sequential	+

* All architectures on the plot apart from Sequential Fourier get as input vector of end effector direction. Because just the fourier features model can not adapt to multimodality results omit direction conditioning at all, just placement of the centroid.

Verification metrics

- Structure is flexible and learnable (Figure 11)

Built prototypes have validated structural rigidity of closed manifolds, even given flexible interconnections between cells. Simulation showed that domain expands exponentially and control loop does not depend on layer length, only computational complexity grows linearly as a function of layers.

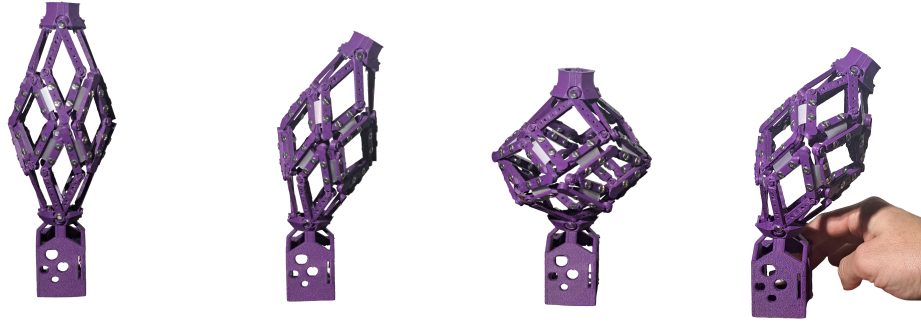


Figure 11 - demonstration of flexibility of the rigid structure

- NiTi positioning accuracy

Using hysteresis loop compensation, here are results obtained by manual measurements of rhombus deformation as a function of displacement sent to the microcontroller. As we can see, the relationship is almost perfect with R^2 fit of 0.92 (see Figure 12), so NiTi can be continuously controlled. Given low quality of observation conditions the results have potential to be improved.

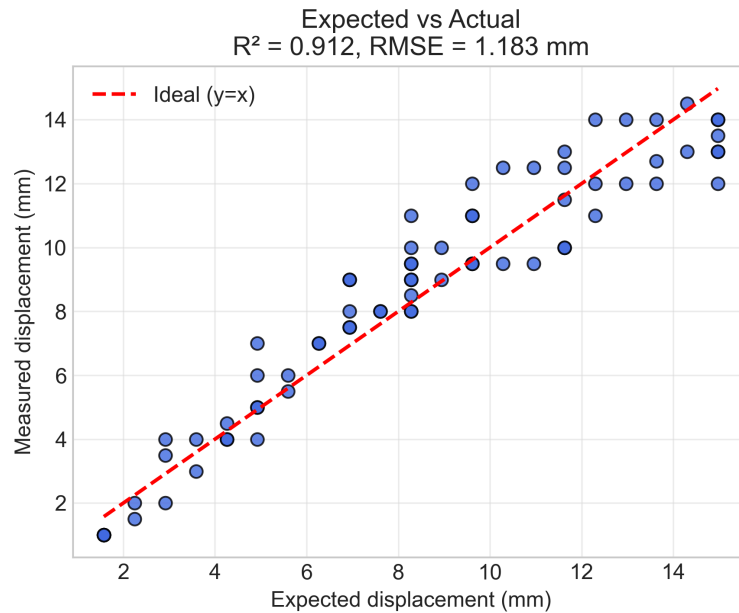


Figure 12 - Evaluation of NiTi voltage regimen to displacement mapping

- Employability on microcontrollers

Sequential architectures showed their compact size that can be quantized and run locally on relatively powerful but hobby level controllers, such as ESP32-S3 [6].

Discussion of results

According to evaluated properties of proposed architecture it is possible to say with confidence that it is flexible and solvable using numerical methods in a relatively short time. It is also very light in terms of memory space and the quantized model can be easily deployed on microcontrollers. However end effector positioning precision reached in these tests is very low and for real world applications, speed and accuracy characteristics should be improved. In addition to that, structures that contain more layers and wider layers should be tested and their flexibility evaluated compared to characteristics found in this project.

The use cases for MEAPER are not limited by a single field or application, MEAPER can be deployed instead of existing soft robotics solutions in cases where speed is required, as a part of a gripper, snake-like mechanisms for rescue teams and others. The control does not depend on the actuation mechanism or length of actuator. So the use of SMA in this case should be questioned in order to improve precision characteristics. In this particular case MEAPER was tested on NiTi, however one can consider servos for some larger scale applications, or even myosin and actin for low scale bio-robotics. In principle once the architecture is chosen, the control can be built by synthesizing a dataset for layer to layer dynamics, edge cases and then a proposed pipeline can be used to control its motion and then preferable actuation material placed into the cellular structure.

Conclusions

- NiTi can achieve repeatable and predictable contraction and can be used in an open control loop if the application can tolerate bounded error.
- Neural networks can provide accurate initial guesses for ambiguous over constrained inverse kinematics problems with many degrees of freedom.
- MEAPER architecture is linearly scalable both in terms of computational complexity and geometric design while providing a rich end-effector workspace.

MEAPER establishes a generalizable, hardware-agnostic control and structural framework. The system successfully demonstrates that soft-actuator kinematics can be emulated using modular rigid-elastic units without sacrificing numerical solver efficiency or embedded controller compatibility. For applications where rapid power delivery and spatial flexibility outweigh fine sub-millimeter precision—such as search-and-rescue mechanisms or adaptive soft grippers—the proposed architecture offers a lightweight, scalable solution.

Future work

1. As was already stated in the discussion section, the actuation material probably should be reconsidered. In addition to that the structure itself should be evaluated on a wider variety of parameters.
2. Beside that real world application will not suffice with open loop control so probably some sort of feedback loop is needed to inform about external forces such as accelerometer.
3. Rhombuses state feedback loop can be used for more accurate positioning. For these purposes one can consider using resistance of NiTi itself with some microvolt sensing circuitry. This will also allow us to develop a failure mode policy if one of the elements will crash. This will utilize redundancy of the structure. In that case a more complex training strategy and optimization should be created.

Acknowledgements

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